

Project Report: On-Device Image Inpainting and Editing Tool

Developing a High-Performance Android Application

Team 34: DROPTABLETEAMS

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Problem Statement

The primary challenge is to develop a high-performance image inpainting application capable of running entirely on-device.

- **Objective:** To design and implement an Android application for real-time image inpainting.
- **Target Platform:** The application will be specifically optimized for the Qualcomm Innovator's Development Kit (QIDK).
- **Core Focus:**
 - Accelerating model inference using the dedicated Neural Processing Unit (NPU).
 - Benchmarking end-to-end performance (latency, power) across different processors: CPU, GPU, and NPU.
 - Evaluating and comparing the output image quality of various inpainting models.

Motivation

- To systematically leverage and evaluate the full capabilities of the CPU, GPU, and NPU on Qualcomm's latest Snapdragon 8 Gen 3 platform via the QIDK.
- To research, compare, and benchmark various image inpainting models based on two key criteria:
 - Output Image Quality
 - On-device Performance (Latency, Throughput)
- To investigate the feasibility and potential of deploying these advanced generative AI models efficiently on resource-constrained embedded systems.

Methodology: Models

Model Selection

- **Primary Model: Lama-Dilated**, a high-performance, dilated version of the original Lama model.

Lama-Dilated: Technical Details

- **Checkpoint:** Dilated CelebAHQ
- **Input Resolution:** 512x512
- **Parameters:** 45.6M
- **Model Size (FP32):** 174 MB

Methodology: Frameworks

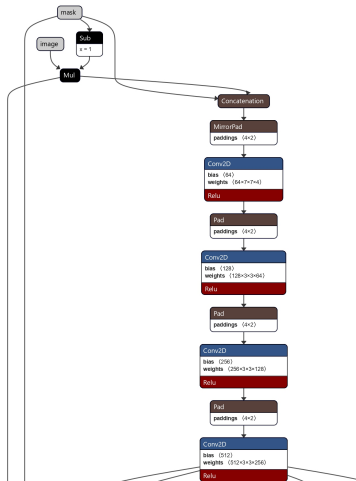
In-App Deployment (Android)

- **Backend:** TensorFlow Lite (TFLite)
- **Integration:** Model is natively integrated into the Kotlin app.
- **Delegates Used:**
 - **NPU:** QNN Delegate
 - **GPU:** TFLite GPU Delegate
 - **CPU:** XNNPack Delegate

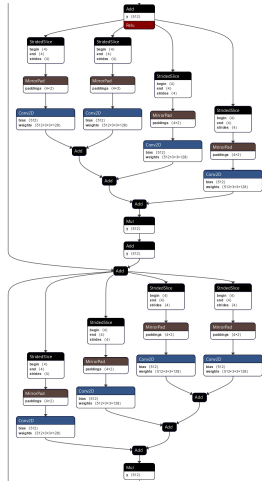
Benchmarking Script (Direct QIDK)

- Used for batch testing (110 images) directly on the QIDK using qnn-net-run binary from QNN SDK, bypassing the app UI using ADB shell.
- **Model Formats:**
 - CPU/GPU: Shared Library (.so)
 - NPU: QNN Context Binary (Compiled from FP32 $\rightarrow$ FP16)

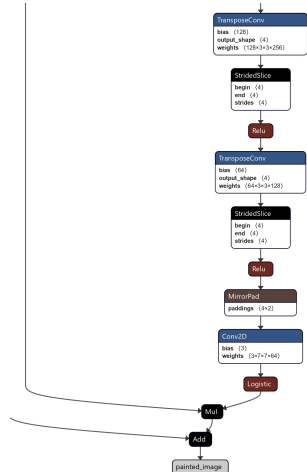
Methodology : Model Internal Block Diagram



Input Layer



Hidden Layer



Output Layer

Note: We have 6 of these hidden layers one after the other between input and output layer.

Methodology: Block Diagram



Diagram showing app workflow and delegate dispatch.

Methodology: Metrics Overview

Strategy: No single metric is perfect. A combination provides a comprehensive understanding of model quality.

Full-Reference Metrics

- PSNR
- SSIM
- LPIPS

No-Reference Metrics

- NIQE
- BRISQUE
- PIQE

Combined Scores

- Combined Full-Reference Score
- Combined Masked Score
- Combined No-Reference Score

Methodology: Parameters Considered (Full-Reference)

- **PSNR (Peak Signal-to-Noise Ratio)**

- **What it is:** A simple metric for pixel-level accuracy.
- **Formula:** $\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right)$
- **Pro:** Measures raw pixel-level reconstruction accuracy.
- **Con:** Does NOT account for perceptual quality.
- **Higher value is better.**

- **SSIM (Structural Similarity Index)**

- **What it is:** Compares local patterns of pixel intensities.
- **How it works:** Computed using luminance, contrast, and structural similarity terms.
- **Pro:** Good for evaluating preservation of textures and edges.
- **Con:** Unreliable for large inpainted regions.
- **Closer value to 1 is better.**

Methodology: Parameters Considered (Full-Reference)

- **LPIPS (Learned Perceptual Image Patch Similarity)**

- **What it is:** Uses a DNN to compare features, closer to human perception.
- **Pro:** Aligns well with human judgment.
- **Con:** Sensitive to small artifacts.
- **Lower value is better.**

Methodology: Parameters Considered (No-Reference)

- **NIQE (Natural Image Quality Evaluator)**

- **What it is:** Evaluates deviation from natural scene statistics.
- **Pro:** Fully no-reference; no training on distorted images needed.
- **Con:** May not align perfectly with human perception in all contexts.
- **Lower value is better.**

- **BRISQUE (Blind/Referenceless IQA)**

- **What it is:** A machine learning-based no-reference metric.
- **Pro:** Correlates well with human opinion; sensitive to common distortions.
- **Con:** Requires model training on human opinion datasets.
- **Lower value is better.**

- **PIQE (Perception-based IQA)**

- **What it is:** A fast, fully blind metric using block-based distortion.
- **Pro:** Computationally efficient, no training required.
- **Con:** May oversimplify complex distortions.
- **Lower value is better.**

Methodology: Parameters Considered (Combined Scores)

Goal: To distill multiple metrics into single, interpretable scores for a clear comparison.

- **Combined Full-Reference Score**

- **Purpose:** Balances pixel/structural accuracy with perceptual quality.
- **Formula:** $0.4 \cdot \text{norm}(\text{PSNR}) + 0.4 \cdot \text{norm}(\text{SSIM}) + 0.2 \cdot \text{norm}_{\text{inv}}(\text{LPIPS})$
- **Justification:** PSNR/SSIM form the core (80%), with LPIPS adding a perceptual component (20%).

- **Combined Masked Score**

- **Purpose:** Focuses evaluation only on the inpainted region.
- **Formula:** $\text{mean}(\text{norm}(\text{M-PSNR}), \text{norm}(\text{M-SSIM}))$
- **Justification:** Equal importance for pixel and structural accuracy within the critical masked area.

- **Combined No-Reference Score**

- **Purpose:** Measures overall "naturalness" of the output.
- **Formula:** $\text{mean}(\text{norm}_{\text{inv}}(\text{NIQE}, \text{BRISQUE}, \text{PIQE}))$
- **Justification:** A robust perceptual score from three distinct blind quality algorithms.

Methodology: Dataset

The Need for Ground Truth Full-reference metrics (PSNR, SSIM) require a "perfect" ground truth image to compare against. This allows for precise, objective evaluation.

- We also need the exact masked region to evaluate masked-PSNR (M-PSNR) and masked-SSIM (M-SSIM).

Our Solution: A Controlled, Custom Dataset

- **Custom Generation:** We algorithmically generated a dataset by placing diverse objects (in various orientations) randomly onto background images. This includes single and multi-object composites.
- **Ground Truth Masks:** This process allows us to create precise single and multi-object masks, enabling focused analysis on the inpainted region.
- **SOTA:** We utilize results from the SOTA model *Clipdrop Cleanup* to establish a high-quality benchmark for our model.
- **Systematic Approach:** Our final dataset contains 100 different background-object combinations, all of which are tested across the CPU, GPU, and NPU.

This rigorous setup allows for objective, repeatable, and meaningful evaluation of our model's performance.

Methodology: Data Pre-processing and Post-processing

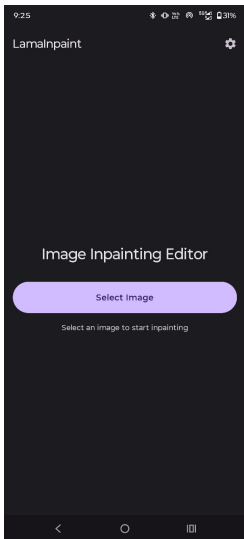
Data Pre-processing

- Convert image and mask to **512x512** resolution.
- Convert the image and mask into **FP32 tensors** for model input.
- Normalize the tensor to $[-1, 1]$ for image and $[0, 1]$ for the mask (black/white tensor).

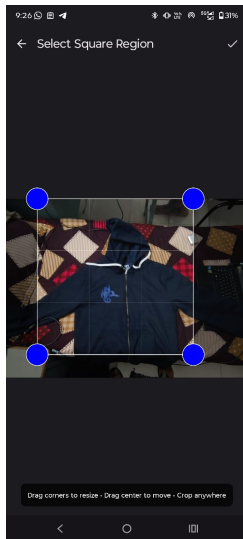
Data Post-processing

- Convert tensor outputted by the model back to image.
- Denormalize the tensor to pixel values $[0, 255]$ for output.

Application: Workflow



1. Main Homepage

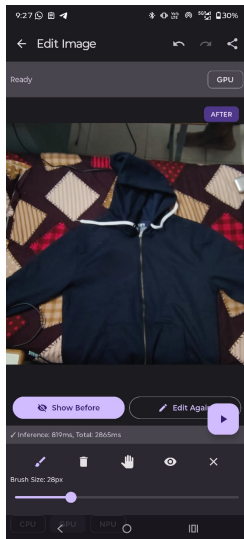


2. Image Cropping

Application: Workflow

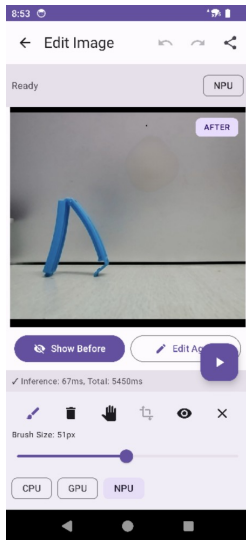


3. Masking Canvas



4. Inpainted Output

Application: Android App Functionalities



Core Functionalities

- Dynamic backend selection (CPU/GPU/NPU) integrated within the app.
- Share the inpainted image directly from the app.

Major UI Upgrades (Mask Editing)

- **We have functionality to select multiple objects in a single mask.**
- Adjustable brush sizes for mask drawing/erasing.
- Full editing controls (undo, redo, clear mask).
- View utilities (hide/show mask, image panning).
- Improved mask visualization and responsiveness.

Model Results: Benchmarking Process

We performed benchmarking in three distinct stages:

① Initial AI-Hub Validation

- Performance was validated on the AI-HUB QIDK simulator and confirmed with initial tests on the physical device.

② Direct QIDK Scripting (Detailed Analysis)

- Ran batch tests (100+ images) directly on QIDK hardware.
- Used for detailed quality (vs. SOTA) and performance (avg. inference) logging.

③ Native App Integration

- Model integrated into the Kotlin app using TFLite delegates.
- Used to profile real-world inference times from within the application.

Quantitative Results: SOTA vs. Our Model

Metric	SOTA (Clipdrop)	Our Model (CPU)	Our Model (GPU)	Our Model (NPU Local)
<i>Full-Reference Metrics (Higher is Better / LPIPS Lower is Better)</i>				
PSNR	38.0713	37.5635	37.5635	36.8460
SSIM	0.9661	0.9788	0.9788	0.9776
LPIPS	0.0175	0.0184	0.0184	0.0185
<i>Masked-Region Metrics (Higher is Better)</i>				
Masked PSNR	28.0377	24.8726	24.8726	24.8167
Masked SSIM	0.7743	0.7737	0.7737	0.7734
<i>Perceptual & No-Reference Metrics (Lower is Better)</i>				
NIQE	6.9108	6.8986	6.8987	6.8581
BRISQUE	39.4363	39.6092	39.6097	38.3802
PIQE	42.4179	41.7735	41.7735	41.7512

Table: Average metric scores across the entire test dataset. Best performance in each category is in bold.

We have attached the complete benchmarking of all 9 metrics - best and worst 3 images across all 100 images at the end.

Analysis of Results & Key Insights

- **Comparable to SOTA:** CPU/GPU models achieve SOTA-level structural (SSIM) and masked-region performance.
- **Perceptual Quality:** SOTA leads slightly in LPIPS, but our models remain highly competitive, preserving visual fidelity.
- **NPU Performance:** NPU (quantized) results are very close to CPU/GPU and even **exceed** them in no-reference metrics (NIQE, BRISQUE).
- **Inference Efficiency:** NPU offers drastic speed and efficiency gains over CPU/GPU, making it ideal for on-device deployment.
- **Masked Region Robustness:** All delegates (CPU, GPU, NPU) maintain strong M-PSNR and M-SSIM scores, ensuring reliable reconstruction.

Model Results: On-Device Inference Latency

TFLite Delegate Performance (In-App)

- **CPU (XNNPack):** ~6800 ms
- **GPU Delegate:** ~253 ms (vs. ~1000 ms earlier)
- **NPU (QNN Delegate):** ~70 ms (vs. ~330 ms earlier)

Key Insights

- On-device inference times show significant improvement when running via the app.
- Optimized model graphs are cached, reducing latency on subsequent runs.
- NPU inference achieves the fastest response while maintaining quality.

Quantization Experiments

- INT8 quantization caused quality degradation.
- Final deployment uses FP16 weights on NPU and GPU, whereas the CPU runs on FP32 — this led to balanced performance and quality.
- Mixed-precision (W8A16) failed due to precision loss.

Model Results: On-Device Inference Latency

QNN Net Run Logs (Scripted)

CPU Log

```
Execute Stats (Overall):
-----
NetRun IPS (includes IO and misc. time): 0.4036 inf/sec

Execute Stats (Average):
-----
Total Inference Time:
-----
Graph 0 (graph_s0tuzqu.):
NetRun: 2474928 us
Backend (GRAPH_EXECUTE): 2473548 us

Execute Stats (Min):
-----
Total Inference Time:
-----
Graph 0 (graph_s0tuzqu.):
NetRun: 2474928 us
Backend (GRAPH_EXECUTE): 2473548 us

Execute Stats (Max):
-----
Total Inference Time:
-----
Graph 0 (graph_s0tuzqu.):
NetRun: 2474928 us
Backend (GRAPH_EXECUTE): 2473548 us

Command executed successfully: ./qnn-profile-viewer --input_log-output/qnn-profiling-data-0.log
--output=bench_csv/0113_tabletnwall_car_book_outputs.csv
Inference time: 2.477523 seconds per inference
Saved visualized output to: bench/0113_tabletnwall_car_book_inpainted.png

Average inference time: 2.332555 seconds per inference
Script finished.
```

CPU (FP32): 2.33 s

GPU Log

```
De-Init Stats:
-----
NetRun: 40624 us
Backend (QnnContext_free): 40620 us

Execute Stats (Overall):
-----
NetRun IPS (includes IO and misc. time): 0.9475 inf/sec

Execute Stats (Average):
-----
Total Inference Time:
-----
Graph 0 (graph_s0tuzqu.):
NetRun: 1046890 us
Backend (QnnGraph_execute): 1046807 us

Execute Stats (Min):
-----
Total Inference Time:
-----
Graph 0 (graph_s0tuzqu.):
NetRun: 1046890 us
Backend (QnnGraph_execute): 1046807 us

Execute Stats (Max):
-----
Total Inference Time:
-----
Graph 0 (graph_s0tuzqu.):
NetRun: 1046890 us
Backend (QnnGraph_execute): 1046807 us

Command executed successfully: ./qnn-profile-viewer --input_log-output/qnn-profiling-data-0.log
--output=bench_csv/0113_tabletnwall_car_book_outputs.csv
Inference time: 1.05427 seconds per inference
Saved visualized output to: bench/0113_tabletnwall_car_book_inpainted.png

Average inference time: 1.052044 seconds per inference
Script finished.
```

GPU (FP32): 1.05 s

NPU Log

```
1 input log file location: output/qnn-profiling-data-0.log
2 qnn-profile-viewer version: v2.38.0.25090114052.125120
3 qnn-net-run version: v2.40.0.251030114326.189385
4 Backend version: v2.40.0.251030114326.189385
5 Msg Timestamp, Message, Time, Unit of Measurement, Timing Source, Event Level, Event Identifier
6 21537065951.INIT.33255.US.NETRUN.ROOT.null
7 21537065951.INIT.26291.US.BACKEND.ROOT.RPC (load binary) time
8 21537065951.INIT.25944.US.BACKEND.ROOT.QNN accelerator (load binary) time
9 21537065951.INIT.25439.US.BACKEND.ROOT.Accelerator (load binary) time
10 21537065951.INIT.33235.US.BACKEND.ROOT.QNN (load binary) time
11 0.COMPOSE.GRAPH5.0.US.NETRUN.ROOT.null
12 21537215703.DE-INIT.3371.US.NETRUN.ROOT.null
13 21537215703.DE-INIT.1146.US.BACKEND.ROOT.RPC (deinit) time
14 21537215703.DE-INIT.942.US.BACKEND.ROOT.QNN Accelerator (deinit) time
15 21537215703.DE-INIT.918.US.BACKEND.ROOT.Accelerator (deinit) time
16 21537215703.DE-INIT.3362.US.BACKEND.ROOT.QNN (deinit) time
17 21537141481.EXECUTE.64948.US.NETRUN.ROOT.Graph 0: graph_bjir5tdv
18 21537141481.EXECUTE.4.ACOUNT.BACKEND.ROOT.Number of MAX threads used
19 21537141481.EXECUTE.61195.US.BACKEND.ROOT.RPC (execute) time
20 21537141481.EXECUTE.60762.US.BACKEND.ROOT.QNN accelerator (execute) time
21 21537141481.EXECUTE.60712.US.BACKEND.ROOT.Accelerator (execute) time
22 21537141481.EXECUTE.60803.US.BACKEND.ROOT.Accelerator (execute excluding wait) time
23 21537141481.EXECUTE.64911.US.BACKEND.ROOT.QNN (execute) time
24 21537211041.EXECUTE IPS: 15.532166, INF/SEC, NETRUN, ROOT, null
```

NPU (FP16): 68 ms

The initial NPU results appeared slower because `qnn-net-run` reports the *entire* processing pipeline time rather than only the inference time, adding significant overhead.

After switching the QNN configuration to **Burst Mode**, the actual inference time improved drastically:

Model Results : AI-hub inference times

Inference Metrics	
Maximum Inference Time ⓘ	Estimated Peak Memory Usage ⓘ
983.8 ms	2 - 33 MB
Runtime Configuration	
QNN Options	
	Value
context_options.htp_options.performance_mode	BURST
default_graph_options.htp_options.optimizations[0].type	FINALIZE_OPTIMIZATION_FLAG
default_graph_options.htp_options.optimizations[0].value	3.0
default_graph_options.htp_options.precision	FP16T16
default_graph_options.htp_options.vtcm_size	8

GPU Inference Time Log

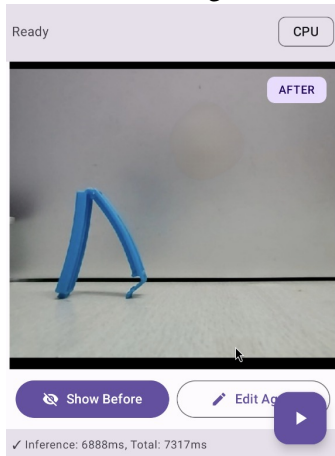
Inference Metrics	
Minimum Inference Time ⓘ	Estimated Peak Memory Usage ⓘ
55.5 ms	91 - 110 MB
Runtime Configuration	
QNN Options	
	Value
context_options.htp_options.performance_mode	BURST
default_graph_options.htp_options.optimizations[0].type	FINALIZE_OPTIMIZATION_FLAG
default_graph_options.htp_options.optimizations[0].value	3.0
default_graph_options.htp_options.precision	FP16T16
default_graph_options.htp_options.vtcm_size	8

NPU Inference Time Log — This gives us a theoretical maximum for the inference times for the model to run. We succeeded in achieving almost identical inference times on the device on next slide.

Model Results: On-Device Inference Latency

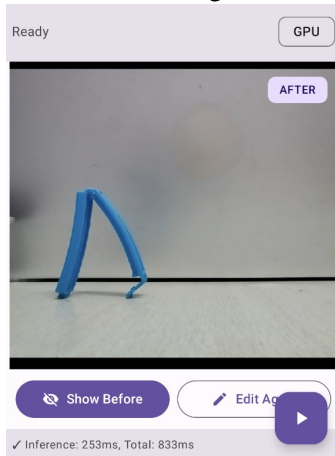
TFLite App Logs (On-Device)

CPU Log



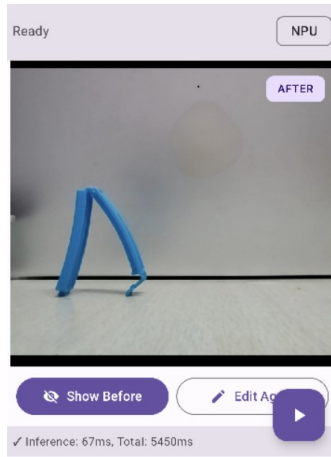
CPU: ~ 6800ms

GPU Log



GPU: ~ 250ms

NPU Log



NPU: ~ 70ms

Logs on different backends

```
0001_grass_bench_outputs.csv X
input log file location, output/qnn-profiling-data_0.log
qnn-profile-viewer version, v2.38.0.250901140452.125126
qnn-net-run version, v2.38.0.250901140452.125126
Backend version, v2.38.0.250901140452.125126
Msg Timestamp, Message, Time, Unit of Measurement, Timing Source, Event Level, Event Identifier
70764603,INIT,214862,US,NETRUN,ROOT,null
70724299,COMPOSE_GRAPH,215190,US,NETRUN,ROOT,null
707265339,FINALIZE_GRAPH,US,NETRUN,ROOT,Graph 0: graph_sbtuzq...
707265339,FINALIZE_GRAPH,US,BACKEND,ROOT,GRAPH_FINALIZE
70903244,DE-INIT,21524,US,NETRUN,ROOT,null
709052445,DE-INIT,0,US,BACKEND,ROOT,null
709018078,EXECUTE,1765376,US,NETRUN,ROOT,Graph 0: graph_sbtuzq...
709018078,EXECUTE,1765376,US,BACKEND,ROOT,GRAPH_EXECUTE
709015005,EXECUTE_IPS,0.572372,1MF/SEC,NETRUN,ROOT,null
```

CPU (FP32)

```
0001_grass_bench_outputs.csv X
gpu_logs > bench.csv > 0001_grass_bench_outputs.csv
input log file location, output/qnn-profiling-data_0.log
qnn-profile-viewer version, v2.38.0.250901140452.125126
qnn-net-run version, v2.38.0.250901140452.125126
Backend version, v2.38.0.250901140452.125126
Msg Timestamp, Message, Time, Unit of Measurement, Timing Source, Event Level
1481257331,INIT,714809,US,NETRUN,ROOT,null
1481383314,COMPOSE_GRAPH,125465,US,NETRUN,ROOT,null
1481972776,FINALIZE_GRAPH,589330,US,NETRUN,ROOT,Graph 0: graph_sbtuzq...
1481972776,FINALIZE_GRAPH,589326,US,BACKEND,ROOT,qnnGraph_finalize
1483899236,DE-INIT,39893,US,NETRUN,ROOT,null
1483899236,DE-INIT,39889,US,BACKEND,ROOT,QnnContext_free
1483883912,EXECUTE,1029391,US,NETRUN,ROOT,Graph 0: graph_sbtuzq...
1483883912,EXECUTE,1029378,US,BACKEND,ROOT,qnnGraph_execute
14838859138,EXECUTE_IPS,0.969882,1MF/SEC,NETRUN,ROOT,null
```

GPU (FP32)

```
0001_grass_bench_outputs.csv X
npu_logs_best > bench.csv > 0001_grass_bench_outputs.csv > data
input log file location, output/qnn-profiling-data.log
qnn-profile-viewer version, v2.38.0.250901140452.125126
qnn-net-run version, v2.38.0.250901140452.125126
Backend version, v2.38.0.250901140452.125126
Msg Timestamp, Message, Time, Unit of Measurement, Timing Source, Event Level
8886672757,INIT,53289,US,NETRUN,ROOT,null
8886672757,INIT,42614,US,BACKEND,ROOT,RPC (load binary) time
8886672757,INIT,42033,US,BACKEND,ROOT,QNN accelerator (load binary) time
8886672757,INIT,41593,US,BACKEND,ROOT,Accelerator (load binary) time
8886672757,INIT,53140,US,BACKEND,ROOT,QNN (load binary) time
0,COMPOSE_GRAPH,0,US,NETRUN,ROOT,null
8887898182,DE-INIT,10351,US,NETRUN,ROOT,null
8887898182,DE-INIT,2539,US,BACKEND,ROOT,RPC (deinit) time
8887898182,DE-INIT,1427,US,BACKEND,ROOT,QNN Accelerator (deinit) time
8887898182,DE-INIT,1365,US,BACKEND,ROOT,Accelerator (deinit) time
8887898182,DE-INIT,10334,US,BACKEND,ROOT,QNN (deinit) time
8887814369,EXECUTE,331568,US,NETRUN,ROOT,Graph 0: graph_bjir5tdv
8887814369,EXECUTE,4,COUNT,BACKEND,ROOT,Number of HWX threads used
8887814369,EXECUTE,32816,US,BACKEND,ROOT,RPC (execute) time
8887814369,EXECUTE,327339,US,BACKEND,ROOT,QNN accelerator (execute) time
8887814369,EXECUTE,327166,US,BACKEND,ROOT,Accelerator (execute) time
8887814369,EXECUTE,323152,US,BACKEND,ROOT,Accelerator (execute excluding w
8887814369,EXECUTE,331131,US,BACKEND,ROOT,QNN (execute) time
8887887420,EXECUTE_IPS,2.950166,1MF/SEC,NETRUN,ROOT,null
```

NPU (FP16)

Expanded Benchmarking: New Models

To ensure a comprehensive evaluation of on-device inpainting, we extended our benchmarking beyond Lama-Dilated to include two additional high-performance models.

1. AOT-GAN

(Aggregated COntextual Transformations)

- Obtained directly from **Qualcomm AI Hub**.
- Known for handling high-resolution inpainting with complex structures.

2. MI-GAN

(Mobile Inpainting GAN)

- A lightweight model specifically designed for mobile devices by *Picsart AI Research*.
- **Key Feature:** Superior visual quality and texture preservation.

Model 2: AOT-GAN

Technical Specifications

- **Source:** Qualcomm AI Hub
- **Checkpoint:** CelebAHQ
- **Input Resolution:** 512x512
- **Parameters:** 15.2M
- **Model Size (Float):** 58.0 MB

Integration Workflow

- Since AOT-GAN is hosted on QAI Hub, we utilized the **same established workflow** as Lama-Dilated.
- Validated on QIDK via ADB before app integration.

The screenshot displays the 'Information' section of a model page on the Qualcomm AI Hub. It provides details about the model 'aotgan_float', including its target device (QCS8550), creator (ai-hub-support@qti.qualcomm.com), input space (image: float32[1, 512, 512, 3], mask: float32[1, 512, 512, 1]), and options (qairt_version: latest). It also lists the versions of the underlying frameworks: TensorFlow Lite (2.17.0), QAI RT (v2.39.0.250925215840_163802), QNN TFLite Delegate (v2.39.0.250925215840_163802), Android (13 (TP1A.220624.014)), and AI Hub (aihub-2025.10.21.0). Below this, the 'Inference Metrics' section shows an estimated inference time of 161 ms, an estimated peak memory usage of 3 - 30 MB, and compute units of 245.

Information		
Name aotgan_float	Target Device QCS8550 (Proxy) Android 12 Qualcomm® QCS8550	
Creator ai-hub-support@qti.qualcomm.com	Target Model Download model	
Input Space image: float32[1, 512, 512, 3] mask: float32[1, 512, 512, 1]	Completion Time 10/26/2025, 2:27:27 AM	
Options qairt_version: latest		
Versions TensorFlow Lite: 2.17.0 QAI RT: v2.39.0.250925215840_163802 QNN TFLite Delegate: v2.39.0.250925215840_163802 Android: 13 (TP1A.220624.014) AI Hub: aihub-2025.10.21.0		

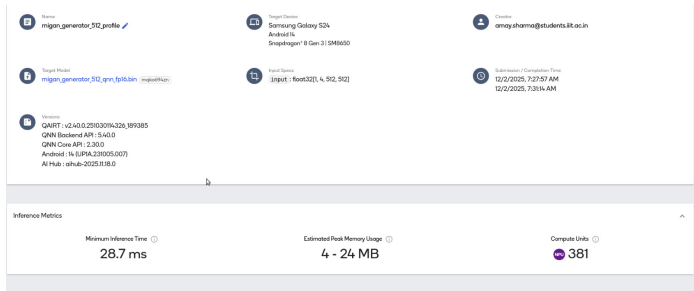
Inference Metrics		
Estimated Inference Time ⓘ	Estimated Peak Memory Usage ⓘ	Compute Units ⓘ
161 ms	3 - 30 MB	245

AOT-GAN Features

Model 3: MI-GAN (Mobile Inpainting GAN)

Overview & Capabilities

- **Source:** Picsart AI Research (GitHub)
- **Parameters:** ~5.9M (Lightweight)
- **Key Strength:** Delivers the **best visual quality** among tested models, with superior texture synthesis.



QAI Hub Compilation & Optimization Log

The Deployment Challenge

Unlike Lama and AOT-GAN, MI-GAN was **not available on QAI Hub**.

- **Issue:** Pre-trained ONNX models failed to initialize on the NPU delegate.
- **Solution:** We built the model from **PyTorch source code** and manually compiled it to ONNX and used Qualcomm AI Hub to compile it to Qualcomm-specific binaries.

Model Results: On-Device Inference Latency

Mi-GAN – QNN Net Run Logs (NPU Only)

Mi-GAN NPU Log

```
mi-gan.npu > qnn-net-run --output-dir . --log-dir . --log qnn-profile-viewer-data-0.log
1  Input log file location, output/qnn-profile-viewer-data-0.log
2  qnn-profile-viewer version, v2.30.0.250901140452 125126
3  qnn-net-run version, v2.40.0.251030114326 169385
4  Backend version, v2.40.0.251030114326 169385
5  Req Timestamp, Message, Time, Unit of Measurement, Timing Source, Event Level, Event Identifier
6  27182236126, INIT, 42687, US, NETRUN, ROOT, null
7  27182236126, INIT, 39079, US, BACKEND, ROOT, RPC (load binary) time
8  27182236126, INIT, 39467, US, BACKEND, ROOT, QNN accelerator (load binary) time
9  27182236126, INIT, 39133, US, BACKEND, ROOT, Accelerator (load binary) time
10 27182236126, INIT, 42643, US, BACKEND, ROOT, QNN (load binary) time
11 0, COMPOSE GRAPHS, 0, US, NETRUN, ROOT, null
12 27182350965, DE-INIT, 3108, US, NETRUN, ROOT, null
13 27182350965, DE-INIT, 1108, US, BACKEND, ROOT, RPC (deinit) time
14 27182350965, DE-INIT, 546, US, BACKEND, ROOT, QNN Accelerator (deinit) time
15 27182350965, DE-INIT, 519, US, BACKEND, ROOT, Accelerator (deinit) time
16 27182350965, DE-INIT, 3095, US, BACKEND, ROOT, QNN (deinit) time
17 27182279187, EXECUTE, 32044, US, NETRUN, ROOT, Graph 0: graph_ymoutc46
18 27182279187, EXECUTE, 4, COUNT, BACKEND, ROOT, Number of HW threads used
19 27182279187, EXECUTE, 3006, US, BACKEND, ROOT, RPC (execute) time
20 27182279187, EXECUTE, 29615, US, BACKEND, ROOT, QNN accelerator (execute) time
21 27182279187, EXECUTE, 29548, US, BACKEND, ROOT, Accelerator (execute) time
22 27182279187, EXECUTE, 29288, US, BACKEND, ROOT, Accelerator (execute excluding wait) time
23 27182279187, EXECUTE, 32014, US, BACKEND, ROOT, QNN (execute) time
24 27182347513, EXECUTE, 195, 24 500000, US, NETRUN, ROOT, null
```

Mi-GAN NPU (FP16): 30 ms

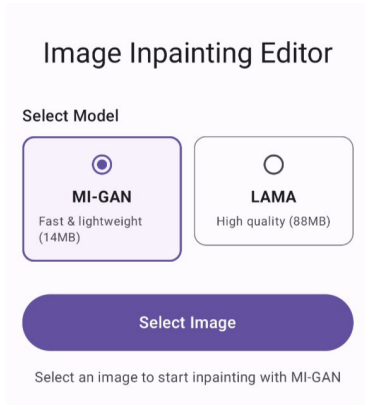
Mi-GAN was benchmarked on-device using `qnn-net-run` in **NPU FP16 mode**. The raw runtime includes the full QNN processing pipeline.

After enabling **Burst Mode**, the pure inference latency is:

Mi-GAN NPU (Burst Mode): 30 ms

MI-GAN: App Integration

Given the superior visual fidelity of MI-GAN, we upgraded the Android application to support **Multi-Model Selection**.



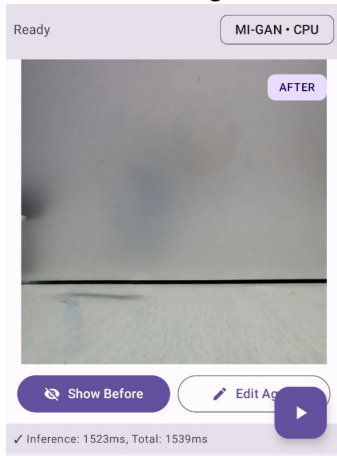
User Experience Upgrades:

- Users can now toggle between **Lama** (Speed focus) and **MI-GAN** (Quality focus) in real-time.
- Both models support dynamic backend switching (CPU ↔ GPU ↔ NPU).

MI-GAN: On-Device Inference Latency

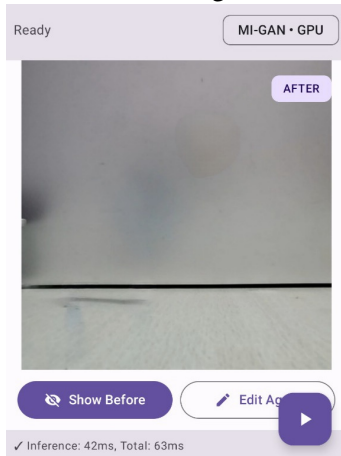
TFLite App Logs (On-Device)

CPU Log



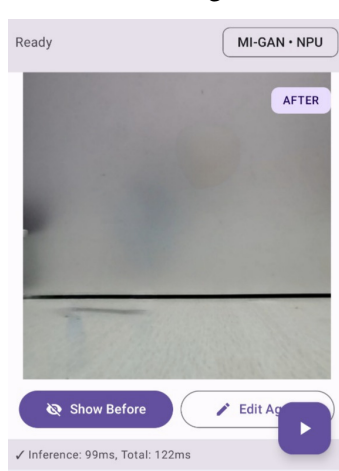
CPU: ~1500 ms

GPU Log



GPU: ~60 ms

NPU Log



NPU: ~100 ms

Quantitative Results: Comparison

Metric	MIGAN (NPU)	MIGAN (GPU)	AOT-GAN (NPU)	LaMa (NPU)
<i>Full-Reference Metrics (Higher is Better / LPIPS Lower is Better)</i>				
PSNR	35.1265	35.1328	34.4822	36.7151
SSIM	0.9491	0.9491	0.9662	0.9779
LPIPS	0.0204	0.0204	0.0214	0.0186
<i>Masked-Region Metrics (Higher is Better)</i>				
Masked PSNR	27.8428	27.8871	25.5917	24.6554
Masked SSIM	0.7685	0.7685	0.7822	0.7766
<i>Perceptual & No-Reference Metrics (Lower is Better)</i>				
NIQE	7.7373	7.5256	5.7254	6.7428
BRISQUE	41.5752	41.5878	37.1004	38.4588
PIQE	50.5731	50.1701	41.9427	41.7638

Table: Comparison benchmarking results across models (best values in bold).

Cross-Model Performance Comparison

A direct comparison of inference latency across all three integrated models. (*Lower is Better*)

Model	CPU Latency	GPU Latency	NPU Latency
Lama-Dilated	~6800 ms	~250 ms	~ 80 ms
AOT-GAN	~3570 ms	~1400 ms	~100 ms
MI-GAN	~1500 ms	~ 60 ms	~100 ms

Insights:

- **MI-GAN** is significantly faster on CPU compared to Lama, making it a viable fallback for older devices.
- **Lama-Dilated** remains the fastest option on the NPU (~80ms).
- **NPU Acceleration** provides a **15x - 20x speedup** compared to CPU execution across all models.
- **Better GPU performance** using MI-GAN as compared to NPU; some model operations are unsupported on the NPU (when compiled to binary, it optimises for Qualcomm specific hardware), so it falls back to CPU, hence the marginally slower NPU inference time.

Thank You

Questions?

Comprehensive Benchmarking Report: CPU, GPU, NPU on LlamaDilated and SOTA , MiGAN-NPU,GPU and AOT-GAN

December 2, 2025

Quantitative Results: SOTA vs. Our Model

Metric	SOTA (Clipdrop)	Our Model (CPU)	Our Model (GPU)	Our Model (NPU Local)
<i>Full-Reference Metrics (Higher is Better / LPIPS Lower is Better)</i>				
PSNR	38.0713	37.5635	37.5635	36.8460
SSIM	0.9661	0.9788	0.9788	0.9776
LPIPS	0.0175	0.0184	0.0184	0.0185
<i>Masked-Region Metrics (Higher is Better)</i>				
Masked PSNR	28.0377	24.8726	24.8726	24.8167
Masked SSIM	0.7743	0.7737	0.7737	0.7734
<i>Perceptual & No-Reference Metrics (Lower is Better)</i>				
NIQE	6.9108	6.8986	6.8987	6.8581
BRISQUE	39.4363	39.6092	39.6097	38.3802
PIQE	42.4179	41.7735	41.7735	41.7512

Table 1: Average metric scores across the entire test dataset. Best performance is in bold.

Metric	MIGAN (NPU)	MIGAN (GPU)	AOT-GAN (NPU)	LaMa (NPU)
<i>Full-Reference Metrics (Higher is Better / LPIPS Lower is Better)</i>				
PSNR	35.1265	35.1328	34.4822	36.7151
SSIM	0.9491	0.9491	0.9662	0.9779
LPIPS	0.0204	0.0204	0.0214	0.0186
<i>Masked-Region Metrics (Higher is Better)</i>				
Masked PSNR	27.8428	27.8871	25.5917	24.6554
Masked SSIM	0.7685	0.7685	0.7822	0.7766
<i>Perceptual & No-Reference Metrics (Lower is Better)</i>				
NIQE	7.7373	7.5256	5.7254	6.7428
BRISQUE	41.5752	41.5878	37.1004	38.4588
PIQE	50.5731	50.1701	41.9427	41.7638

Table 2: Average metric scores across the entire test dataset. Best performance is in bold.

Benchmarking Report: Benchmarking Result SOTA

Full-Reference Metrics - Averages

METRIC	AVERAGE
PSNR	38.0713
SSIM	0.9661
LPIPS	0.0175

Combined Full-Reference (formula): $0.4 \cdot \text{norm}(\text{PSNR}) + 0.4 \cdot \text{norm}(\text{SSIM}) + 0.2 \cdot \text{norm_inv}(\text{LPIPS})$

Best Combined

No	INP_FILE	COMBINED_FULLREF
28	0028_wall-floor_plane_inpainted.jpg	1.0000
11	0011_empty_room_giraffe_inpainted.jpg	0.9919
54	0054_tabletop_book_inpainted.jpg	0.9763

Worst Combined

No	INP_FILE	COMBINED_FULLREF
57	0057_playground_bench_inpainted.jpg	0.0000
41	0041_street_bench_inpainted.jpg	0.0797
73	0073_brick_bench_inpainted.jpg	0.1878

Masked-Region Metrics - Averages

METRIC	AVERAGE
masked PSNR	28.0377
masked SSIM	0.7743

Combined Masked (formula): $\text{mean}(\text{norm}(\text{Masked PSNR}), \text{norm}(\text{Masked SSIM}))$

Best Combined

No	INP_FILE	COMBINED_MASKED
28	0028_wall-floor_plane_inpainted.jpg	1.0000
54	0054_tabletop_book_inpainted.jpg	0.9931
11	0011_empty_room_giraffe_inpainted.jpg	0.9877

Worst Combined

No	INP_FILE	COMBINED_MASKED
57	0057_playground_bench_inpainted.jpg	0.0000
64	0064_playground_mug_inpainted.jpg	0.0302
61	0061_playground_car_inpainted.jpg	0.0546

No-Reference Metrics - Averages

METRIC	AVERAGE
niqe	6.9108
brisque	39.4363
piqe	42.4179

Combined No-Reference (formula): $\text{mean}(\text{norm_inv}(\text{NIQE}), \text{norm_inv}(\text{BRISQUE}), \text{norm_inv}(\text{PIQE}))$

Best Combined

No	INP_FILE	COMBINED_NOREF
17	0017_tablenwall_bench_inpainted.jpg	0.9709
18	0018_tablenwall_tree_inpainted.jpg	0.9684
23	0023_tablenwall_ronaldo_inpainted.jpg	0.9665

Worst Combined

No	INP_FILE	COMBINED_NOREF
36	0036_sunset_plane_inpainted.jpg	0.2396
39	0039_sunset_ronaldo_inpainted.jpg	0.3474
34	0034_sunset_tree_inpainted.jpg	0.3558

Top/Bottom Images per Metric

Full-Reference

PSNR (Higher is better)

Best			Worst		
No	INP_FILE	PSNR	No	INP_FILE	PSNR
28	0028_wall-floor_plane_inpainted.jpg	56.3300	57	0057_playground_bench_inpainted.jpg	18.4600
11	0011_empty_room_giraffe_inpainted.jpg	55.8300	64	0064_playground_mug_inpainted.jpg	20.3200
54	0054_tabletop_book_inpainted.jpg	54.2600	65	0065_beach_bench_inpainted.jpg	21.2300

SSIM (Higher is better)

Best			Worst		
No	INP_FILE	SSIM	No	INP_FILE	SSIM
28	0028_wall-floor_plane_inpainted.jpg	0.9994	57	0057_playground_bench_inpainted.jpg	0.8441
27	0027_wall-floor_giraffe_inpainted.jpg	0.9994	41	0041_street_bench_inpainted.jpg	0.8472
26	0026_wall-floor_tree_inpainted.jpg	0.9993	73	0073_brick_bench_inpainted.jpg	0.8649

LPIPS (Lower is better)

Best			Worst		
No	INP_FILE	LPIPS	No	INP_FILE	LPIPS
26	0026_wall-floor_tree_inpainted.jpg	0.0001	57	0057_playground_bench_inpainted.jpg	0.0954
27	0027_wall-floor_giraffe_inpainted.jpg	0.0001	41	0041_street_bench_inpainted.jpg	0.0778
28	0028_wall-floor_plane_inpainted.jpg	0.0001	70	0070_beach_book_inpainted.jpg	0.0777

Masked-Region

MASKED_PSNR (Higher is better)

Best		
No	INP_FILE	MASKED_PSNR
28	0028_wall-floor_plane_inpainted.jpg	45.0700
54	0054_tabletop_book_inpainted.jpg	44.8100
11	0011_empty_room_giraffe_inpainted.jpg	44.6700
Worst		
No	INP_FILE	MASKED_PSNR
57	0057_playground_bench_inpainted.jpg	11.3100
64	0064_playground_mug_inpainted.jpg	11.3900
58	0058_playground_tree_inpainted.jpg	12.1900

MASKED_SSIM (Higher is better)

Best		
No	INP_FILE	MASKED_SSIM
29	0029_wall-floor_car_inpainted.jpg	0.9953
28	0028_wall-floor_plane_inpainted.jpg	0.9953
26	0026_wall-floor_tree_inpainted.jpg	0.9947
Worst		
No	INP_FILE	MASKED_SSIM
57	0057_playground_bench_inpainted.jpg	0.3724
61	0061_playground_car_inpainted.jpg	0.3938
64	0064_playground_mug_inpainted.jpg	0.4085

No-Reference

NIQE (Lower is better)

Best			Worst		
No	INP_FILE	NIQE	No	INP_FILE	NIQE
43	0043_street_giraffe_inpainted.jpg	3.6617	36	0036_sunset_plane_inpainted.jpg	24.5726
45	0045_street_car_inpainted.jpg	3.7023	40	0040_sunset_mug_inpainted.jpg	22.7513
48	0048_street_mug_inpainted.jpg	3.7600	39	0039_sunset_ronaldo_inpainted.jpg	20.8341

BRISQUE (Lower is better)

Best			Worst		
No	INP_FILE	BRISQUE	No	INP_FILE	BRISQUE
17	0017_tablewall_bench_inpainted.jpg	0.1009	36	0036_sunset_plane_inpainted.jpg	94.5568
18	0018_tablewall_tree_inpainted.jpg	0.4563	39	0039_sunset_ronaldo_inpainted.jpg	92.8665
23	0023_tablewall_ronaldo_inpainted.jpg	0.5305	34	0034_sunset_tree_inpainted.jpg	91.9145

PIQE (Lower is better)

Best			Worst		
No	INP_FILE	PIQE	No	INP_FILE	PIQE
37	0037_sunset_car_inpainted.jpg	21.5515	28	0028_wall-floor_plane_inpainted.jpg	77.2949
40	0040_sunset_mug_inpainted.jpg	22.3123	29	0029_wall-floor_car_inpainted.jpg	77.1141
35	0035_sunset_giraffe_inpainted.jpg	23.1819	27	0027_wall-floor_giraffe_inpainted.jpg	77.1141

Benchmarking Report: Benchmarking result CPU

Full-Reference Metrics - Averages

METRIC	AVERAGE
PSNR	37.5635
SSIM	0.9788
LPIPS	0.0184

Combined Full-Reference (formula): $0.4 \cdot \text{norm}(\text{PSNR}) + 0.4 \cdot \text{norm}(\text{SSIM}) + 0.2 \cdot \text{norm_inv}(\text{LPIPS})$

Best Combined

No	INP_FILE	COMBINED_FULLREF
20	0020_tablenwall_plane_inpainted.png	0.9969
52	0052_tabletop_plane_inpainted.png	0.9939
56	0056_tabletop_mug_inpainted.png	0.9759

Worst Combined

No	INP_FILE	COMBINED_FULLREF
70	0070_beach_book_inpainted.png	0.0684
73	0073_brick_bench_inpainted.png	0.1098
57	0057_playground_bench_inpainted.png	0.1247

Masked-Region Metrics - Averages

METRIC	AVERAGE
masked PSNR	24.8726
masked SSIM	0.7737

Combined Masked (formula): $\text{mean}(\text{norm}(\text{Masked PSNR}), \text{norm}(\text{Masked SSIM}))$

Best Combined

No	INP_FILE	COMBINED_MASKED
56	0056_tabletop_mug_inpainted.png	1.0000
52	0052_tabletop_plane_inpainted.png	0.9818
26	0026_wall-floor_tree_inpainted.png	0.9766

Worst Combined

No	INP_FILE	COMBINED_MASKED
64	0064_playground_mug_inpainted.png	0.0580
77	0077_brick_car_inpainted.png	0.1075
69	0069_beach_car_inpainted.png	0.1477

No-Reference Metrics - Averages

METRIC	AVERAGE
niqe	6.8986
brisque	39.6092
piqe	41.7735

Combined No-Reference (formula): $\text{mean}(\text{norm_inv}(\text{NIQE}), \text{norm_inv}(\text{BRISQUE}), \text{norm_inv}(\text{PIQE}))$

Best Combined

No	INP_FILE	COMBINED_NOREF
17	0017_tablenwall_bench_inpainted.png	0.9532
23	0023_tablenwall_ronaldo_inpainted.png	0.9472
21	0021_tablenwall_car_inpainted.png	0.9462

Worst Combined

No	INP_FILE	COMBINED_NOREF
36	0036_sunset_plane_inpainted.png	0.1718
39	0039_sunset_ronaldo_inpainted.png	0.2550
40	0040_sunset_mug_inpainted.png	0.3229

Top/Bottom Images per Metric

Full-Reference

PSNR (Higher is better)

Best			Worst		
No	INP_FILE	PSNR	No	INP_FILE	PSNR
20	0020_tablenwall_plane_inpainted.png	54.6300	70	0070_beach_book_inpainted.png	21.2400
52	0052_tabletop_plane_inpainted.png	54.2400	64	0064_playground_mug_inpainted.png	21.5700
56	0056_tabletop_mug_inpainted.png	52.6600	69	0069_beach_car_inpainted.png	21.9600

SSIM (Higher is better)

Best			Worst		
No	INP_FILE	SSIM	No	INP_FILE	SSIM
52	0052_tabletop_plane_inpainted.png	0.9990	73	0073_brick_bench_inpainted.png	0.9166
56	0056_tabletop_mug_inpainted.png	0.9989	74	0074_brick_tree_inpainted.png	0.9227
26	0026_wall-floor_tree_inpainted.png	0.9988	57	0057_playground_bench_inpainted.png	0.9272

LPIPS (Lower is better)

Best			Worst		
No	INP_FILE	LPIPS	No	INP_FILE	LPIPS
56	0056_tabletop_mug_inpainted.png	0.0003	70	0070_beach_book_inpainted.png	0.0721
52	0052_tabletop_plane_inpainted.png	0.0008	73	0073_brick_bench_inpainted.png	0.0576
20	0020_tablenwall_plane_inpainted.png	0.0009	74	0074_brick_tree_inpainted.png	0.0542

Masked-Region

MASKED_PSNR (Higher is better)

Best		
No	INP_FILE	MASKED_PSNR
56	0056_tabletop_mug_inpainted.png	40.4000
52	0052_tabletop_plane_inpainted.png	39.4100
26	0026_wall-floor_tree_inpainted.png	39.0200
Worst		
No	INP_FILE	MASKED_PSNR
64	0064_playground_mug_inpainted.png	9.9900
70	0070_beach_book_inpainted.png	10.7900
58	0058_playground_tree_inpainted.png	11.1100

MASKED_SSIM (Higher is better)

Best		
No	INP_FILE	MASKED_SSIM
56	0056_tabletop_mug_inpainted.png	0.9860
12	0012_empty_room_plane_inpainted.png	0.9856
26	0026_wall-floor_tree_inpainted.png	0.9849
Worst		
No	INP_FILE	MASKED_SSIM
77	0077_brick_car_inpainted.png	0.2599
64	0064_playground_mug_inpainted.png	0.3441
74	0074_brick_tree_inpainted.png	0.3938

No-Reference

NIQE (Lower is better)

Best			Worst		
No	INP_FILE	NIQE	No	INP_FILE	NIQE
48	0048_street_mug_inpainted.png	3.4525	40	0040_sunset_mug_inpainted.png	28.5577
41	0041_street_bench_inpainted.png	3.4860	39	0039_sunset_ronaldo_inpainted.png	28.3273
47	0047_street_ronaldo_inpainted.png	3.5091	36	0036_sunset_plane_inpainted.png	28.0210

BRISQUE (Lower is better)

Best			Worst		
No	INP_FILE	BRISQUE	No	INP_FILE	BRISQUE
24	0024_tablenwall_mug_inpainted.png	0.5572	36	0036_sunset_plane_inpainted.png	103.0840
17	0017_tablenwall_bench_inpainted.png	0.6978	35	0035_sunset_giraffe_inpainted.png	101.4219
18	0018_tablenwall_tree_inpainted.png	0.8411	40	0040_sunset_mug_inpainted.png	101.0929

PIQE (Lower is better)

Best			Worst		
No	INP_FILE	PIQE	No	INP_FILE	PIQE
38	0038_sunset_book_inpainted.png	18.5697	29	0029_wall-floor_car_inpainted.png	74.7471
37	0037_sunset_car_inpainted.png	19.6817	31	0031_wall-floor_ronaldo_inpainted.png	74.2748
40	0040_sunset_mug_inpainted.png	21.4241	26	0026_wall-floor_tree_inpainted.png	73.6430

Benchmarking Report: GPU

Full-Reference Metrics - Averages

METRIC	AVERAGE
PSNR	37.5635
SSIM	0.9788
LPIPS	0.0184

Combined Full-Reference (formula): $0.4 \cdot \text{norm}(\text{PSNR}) + 0.4 \cdot \text{norm}(\text{SSIM}) + 0.2 \cdot \text{norm_inv}(\text{LPIPS})$

Best Combined

No	INP_FILE	COMBINED_FULLREF
20	0020_tablenwall_plane_inpainted.png	0.9969
52	0052_tabletop_plane_inpainted.png	0.9939
56	0056_tabletop_mug_inpainted.png	0.9759

Worst Combined

No	INP_FILE	COMBINED_FULLREF
70	0070_beach_book_inpainted.png	0.0684
73	0073_brick_bench_inpainted.png	0.1098
57	0057_playground_bench_inpainted.png	0.1247

Masked-Region Metrics - Averages

METRIC	AVERAGE
masked PSNR	24.8726
masked SSIM	0.7737

Combined Masked (formula): $\text{mean}(\text{norm}(\text{Masked PSNR}), \text{norm}(\text{Masked SSIM}))$

Best Combined

No	INP_FILE	COMBINED_MASKED
56	0056_tabletop_mug_inpainted.png	1.0000
52	0052_tabletop_plane_inpainted.png	0.9818
26	0026_wall-floor_tree_inpainted.png	0.9766

Worst Combined

No	INP_FILE	COMBINED_MASKED
64	0064_playground_mug_inpainted.png	0.0580
77	0077_brick_car_inpainted.png	0.1075
69	0069_beach_car_inpainted.png	0.1477

No-Reference Metrics - Averages

METRIC	AVERAGE
niqe	6.8987
brisque	39.6097
piqe	41.7735

Combined No-Reference (formula): $\text{mean}(\text{norm_inv}(\text{NIQE}), \text{norm_inv}(\text{BRISQUE}), \text{norm_inv}(\text{PIQE}))$

Best Combined

No	INP_FILE	COMBINED_NOREF
17	0017_tablenwall_bench_inpainted.png	0.9530
23	0023_tablenwall_ronaldo_inpainted.png	0.9472
21	0021_tablenwall_car_inpainted.png	0.9462

Worst Combined

No	INP_FILE	COMBINED_NOREF
36	0036_sunset_plane_inpainted.png	0.1718
39	0039_sunset_ronaldo_inpainted.png	0.2550
40	0040_sunset_mug_inpainted.png	0.3228

Top/Bottom Images per Metric

Full-Reference

PSNR (Higher is better)

Best			Worst		
No	INP_FILE	PSNR	No	INP_FILE	PSNR
20	0020_tablenwall_plane_inpainted.png	54.6300	70	0070_beach_book_inpainted.png	21.2400
52	0052_tabletop_plane_inpainted.png	54.2400	64	0064_playground_mug_inpainted.png	21.5700
56	0056_tabletop_mug_inpainted.png	52.6600	69	0069_beach_car_inpainted.png	21.9600

SSIM (Higher is better)

Best			Worst		
No	INP_FILE	SSIM	No	INP_FILE	SSIM
52	0052_tabletop_plane_inpainted.png	0.9990	73	0073_brick_bench_inpainted.png	0.9166
56	0056_tabletop_mug_inpainted.png	0.9989	74	0074_brick_tree_inpainted.png	0.9227
26	0026_wall-floor_tree_inpainted.png	0.9988	57	0057_playground_bench_inpainted.png	0.9272

LPIPS (Lower is better)

Best			Worst		
No	INP_FILE	LPIPS	No	INP_FILE	LPIPS
56	0056_tabletop_mug_inpainted.png	0.0003	70	0070_beach_book_inpainted.png	0.0721
52	0052_tabletop_plane_inpainted.png	0.0008	73	0073_brick_bench_inpainted.png	0.0576
20	0020_tablenwall_plane_inpainted.png	0.0009	74	0074_brick_tree_inpainted.png	0.0542

Masked-Region

MASKED_PSNR (Higher is better)

Best		
No	INP_FILE	MASKED_PSNR
56	0056_tabletop_mug_inpainted.png	40.4000
52	0052_tabletop_plane_inpainted.png	39.4100
26	0026_wall-floor_tree_inpainted.png	39.0200
Worst		
No	INP_FILE	MASKED_PSNR
64	0064_playground_mug_inpainted.png	9.9900
70	0070_beach_book_inpainted.png	10.7900
58	0058_playground_tree_inpainted.png	11.1100

MASKED_SSIM (Higher is better)

Best		
No	INP_FILE	MASKED_SSIM
56	0056_tabletop_mug_inpainted.png	0.9860
12	0012_empty_room_plane_inpainted.png	0.9856
26	0026_wall-floor_tree_inpainted.png	0.9849
Worst		
No	INP_FILE	MASKED_SSIM
77	0077_brick_car_inpainted.png	0.2599
64	0064_playground_mug_inpainted.png	0.3441
74	0074_brick_tree_inpainted.png	0.3938

No-Reference

NIQE (Lower is better)

Best			Worst		
No	INP_FILE	NIQE	No	INP_FILE	NIQE
48	0048_street_mug_inpainted.png	3.4525	40	0040_sunset_mug_inpainted.png	28.5577
41	0041_street_bench_inpainted.png	3.4861	39	0039_sunset_ronaldo_inpainted.png	28.3273
47	0047_street_ronaldo_inpainted.png	3.5115	36	0036_sunset_plane_inpainted.png	28.0202

BRISQUE (Lower is better)

Best			Worst		
No	INP_FILE	BRISQUE	No	INP_FILE	BRISQUE
24	0024_tablewall_mug_inpainted.png	0.5572	36	0036_sunset_plane_inpainted.png	103.0718
17	0017_tablewall_bench_inpainted.png	0.7714	35	0035_sunset_giraffe_inpainted.png	101.4219
18	0018_tablewall_tree_inpainted.png	0.8472	40	0040_sunset_mug_inpainted.png	101.0929

PIQE (Lower is better)

Best			Worst		
No	INP_FILE	PIQE	No	INP_FILE	PIQE
38	0038_sunset_book_inpainted.png	18.5697	29	0029_wall-floor_car_inpainted.png	74.7475
37	0037_sunset_car_inpainted.png	19.6817	31	0031_wall-floor_ronaldo_inpainted.png	74.2748
40	0040_sunset_mug_inpainted.png	21.4241	26	0026_wall-floor_tree_inpainted.png	73.6433

Benchmarking Report: Benchmarking Result NPU

Full-Reference Metrics - Averages

METRIC	AVERAGE
PSNR	36.8460
SSIM	0.9776
LPIPS	0.0185

Combined Full-Reference (formula): $0.4 \cdot \text{norm}(\text{PSNR}) + 0.4 \cdot \text{norm}(\text{SSIM}) + 0.2 \cdot \text{norm_inv}(\text{LPIPS})$

Best Combined

No	INP_FILE	COMBINED_FULLREF
20	0020_tablenwall_plane_inpainted.png	0.9925
52	0052_tabletop_plane_inpainted.png	0.9897
56	0056_tabletop_mug_inpainted.png	0.9759

Worst Combined

No	INP_FILE	COMBINED_FULLREF
70	0070_beach_book_inpainted.png	0.0488
73	0073_brick_bench_inpainted.png	0.1209
57	0057_playground_bench_inpainted.png	0.1243

Masked-Region Metrics - Averages

METRIC	AVERAGE
masked PSNR	24.8167
masked SSIM	0.7734

Combined Masked (formula): $\text{mean}(\text{norm}(\text{Masked PSNR}), \text{norm}(\text{Masked SSIM}))$

Best Combined

No	INP_FILE	COMBINED_MASKED
56	0056_tabletop_mug_inpainted.png	0.9982
52	0052_tabletop_plane_inpainted.png	0.9971
26	0026_wall-floor_tree_inpainted.png	0.9836

Worst Combined

No	INP_FILE	COMBINED_MASKED
64	0064_playground_mug_inpainted.png	0.0576
77	0077_brick_car_inpainted.png	0.1119
69	0069_beach_car_inpainted.png	0.1476

No-Reference Metrics - Averages

METRIC	AVERAGE
niqe	6.8581
brisque	38.3802
piqe	41.7512

Combined No-Reference (formula): $\text{mean}(\text{norm_inv}(\text{NIQE}), \text{norm_inv}(\text{BRISQUE}), \text{norm_inv}(\text{PIQE}))$

Best Combined

No	INP_FILE	COMBINED_NOREF
17	0017_tablenwall_bench_inpainted.png	0.9415
23	0023_tablenwall_ronaldo_inpainted.png	0.9393
21	0021_tablenwall_car_inpainted.png	0.9337

Worst Combined

No	INP_FILE	COMBINED_NOREF
36	0036_sunset_plane_inpainted.png	0.2055
39	0039_sunset_ronaldo_inpainted.png	0.2976
35	0035_sunset_giraffe_inpainted.png	0.3729

Top/Bottom Images per Metric

Full-Reference

PSNR (Higher is better)

Best			Worst		
No	INP_FILE	PSNR	No	INP_FILE	PSNR
20	0020_tablenwall_plane_inpainted.png	50.2600	70	0070_beach_book_inpainted.png	21.2400
52	0052_tabletop_plane_inpainted.png	49.5900	64	0064_playground_mug_inpainted.png	21.5900
19	0019_tablenwall_giraffe_inpainted.png	49.0500	69	0069_beach_car_inpainted.png	21.9700

SSIM (Higher is better)

Best			Worst		
No	INP_FILE	SSIM	No	INP_FILE	SSIM
52	0052_tabletop_plane_inpainted.png	0.9984	73	0073_brick_bench_inpainted.png	0.9164
26	0026_wall-floor_tree_inpainted.png	0.9982	74	0074_brick_tree_inpainted.png	0.9224
56	0056_tabletop_mug_inpainted.png	0.9982	57	0057_playground_bench_inpainted.png	0.9264

LPIPS (Lower is better)

Best			Worst		
No	INP_FILE	LPIPS	No	INP_FILE	LPIPS
56	0056_tabletop_mug_inpainted.png	0.0005	70	0070_beach_book_inpainted.png	0.0723
52	0052_tabletop_plane_inpainted.png	0.0009	73	0073_brick_bench_inpainted.png	0.0576
20	0020_tablenwall_plane_inpainted.png	0.0016	74	0074_brick_tree_inpainted.png	0.0541

Masked-Region

MASKED_PSNR (Higher is better)

Best		
No	INP_FILE	MASKED_PSNR
52	0052_tabletop_plane_inpainted.png	39.2500
56	0056_tabletop_mug_inpainted.png	39.2200
26	0026_wall-floor_tree_inpainted.png	38.3900
Worst		
No	INP_FILE	MASKED_PSNR
64	0064_playground_mug_inpainted.png	10.0200
70	0070_beach_book_inpainted.png	10.8000
58	0058_playground_tree_inpainted.png	11.1400

MASKED_SSIM (Higher is better)

Best		
No	INP_FILE	MASKED_SSIM
12	0012_empty_room_plane_inpainted.png	0.9870
56	0056_tabletop_mug_inpainted.png	0.9851
10	0010_empty_room_tree_inpainted.png	0.9848
Worst		
No	INP_FILE	MASKED_SSIM
77	0077_brick_car_inpainted.png	0.2602
64	0064_playground_mug_inpainted.png	0.3440
74	0074_brick_tree_inpainted.png	0.3940

No-Reference

NIQE (Lower is better)

Best			Worst		
No	INP_FILE	NIQE	No	INP_FILE	NIQE
48	0048_street_mug_inpainted.png	3.4336	36	0036_sunset_plane_inpainted.png	28.4850
41	0041_street_bench_inpainted.png	3.4789	39	0039_sunset_ronaldo_inpainted.png	25.6728
47	0047_street_ronaldo_inpainted.png	3.5005	35	0035_sunset_giraffe_inpainted.png	23.6159

BRISQUE (Lower is better)

Best			Worst		
No	INP_FILE	BRISQUE	No	INP_FILE	BRISQUE
24	0024_tablenwall_mug_inpainted.png	0.0400	36	0036_sunset_plane_inpainted.png	97.4691
18	0018_tablenwall_tree_inpainted.png	0.1646	35	0035_sunset_giraffe_inpainted.png	96.2219
17	0017_tablenwall_bench_inpainted.png	0.1896	40	0040_sunset_mug_inpainted.png	96.0173

PIQE (Lower is better)

Best			Worst		
No	INP_FILE	PIQE	No	INP_FILE	PIQE
40	0040_sunset_mug_inpainted.png	17.6778	29	0029_wall-floor_car_inpainted.png	74.1948
38	0038_sunset_book_inpainted.png	18.5927	31	0031_wall-floor_ronaldo_inpainted.png	73.3298
34	0034_sunset_tree_inpainted.png	19.6667	28	0028_wall-floor_plane_inpainted.png	71.5545

Comprehensive Benchmarking Report: MIGAN, AOT-GAN, LaMa

December 2, 2025

Quantitative Results: Comparison

Metric	MIGAN (NPU)	MIGAN (GPU)	AOT-GAN (NPU)	LaMa (NPU)
<i>Full-Reference Metrics (Higher is Better / LPIPS Lower is Better)</i>				
PSNR	35.1265	35.1328	34.4822	36.7151
SSIM	0.9491	0.9491	0.9662	0.9779
LPIPS	0.0204	0.0204	0.0214	0.0186
<i>Masked-Region Metrics (Higher is Better)</i>				
Masked PSNR	27.8428	27.8871	25.5917	24.6554
Masked SSIM	0.7685	0.7685	0.7822	0.7766
<i>Perceptual & No-Reference Metrics (Lower is Better)</i>				
NIQE	7.7373	7.5256	5.7254	6.7428
BRISQUE	41.5752	41.5878	37.1004	38.4588
PIQE	50.5731	50.1701	41.9427	41.7638

Table 1: Average metric scores across the entire test dataset. Best performance is in bold.

Benchmarking Report: MIGAN (NPU)

Full-Reference Metrics - Averages

METRIC	AVERAGE
PSNR	35.1265
SSIM	0.9491
LPIPS	0.0204

Combined Full-Reference (formula): $0.4 \cdot \text{norm}(\text{PSNR}) + 0.4 \cdot \text{norm}(\text{SSIM}) + 0.2 \cdot \text{norm_inv}(\text{LPIPS})$

Best Combined

No	INP_FILE	COMBINED_FULLREF
52	0052_tabletop_plane_inpainted.jpg	0.9996
20	0020_tablenwall_plane_inpainted.jpg	0.9991
19	0019_tablenwall_giraffe_inpainted.jpg	0.9972

Worst Combined

No	INP_FILE	COMBINED_FULLREF
63	0063_playground_plane_inpainted.jpg	0.0463
57	0057_playground_bench_inpainted.jpg	0.0820
65	0065_beach_bench_inpainted.jpg	0.1171

Masked-Region Metrics - Averages

METRIC	AVERAGE
masked PSNR	27.8428
masked SSIM	0.7685

Combined Masked (formula): $\text{mean}(\text{norm}(\text{Masked PSNR}), \text{norm}(\text{Masked SSIM}))$

Best Combined

No	INP_FILE	COMBINED_MASKED
52	0052_tabletop_plane_inpainted.jpg	1.0000
56	0056_tabletop_mug_inpainted.jpg	0.9856
20	0020_tablenwall_plane_inpainted.jpg	0.9828

Worst Combined

No	INP_FILE	COMBINED_MASKED
63	0063_playground_plane_inpainted.jpg	0.0381
61	0061_playground_car_inpainted.jpg	0.1086
65	0065_beach_bench_inpainted.jpg	0.1088

No-Reference Metrics - Averages

METRIC	AVERAGE
niqe	7.7373
brisque	41.5752
piqe	50.5731

Combined No-Reference (formula): $\text{mean}(\text{norm_inv}(\text{NIQE}), \text{norm_inv}(\text{BRISQUE}), \text{norm_inv}(\text{PIQE}))$

Best Combined

No	INP_FILE	COMBINED_NOREF
48	0048_street_mug_inpainted.jpg	0.9413
44	0044_street_plane_inpainted.jpg	0.9317
45	0045_street_car_inpainted.jpg	0.9231

Worst Combined

No	INP_FILE	COMBINED_NOREF
35	0035_sunset_giraffe_inpainted.jpg	0.1983
36	0036_sunset_plane_inpainted.jpg	0.2057
40	0040_sunset_mug_inpainted.jpg	0.2520

Top/Bottom Images per Metric

Full-Reference

PSNR (Higher is better)

Best			Worst		
No	INP_FILE	PSNR	No	INP_FILE	PSNR
52	0052_tabletop_plane_inpainted.jpg	50.9100	63	0063_playground_plane_inpainted.jpg	17.5800
20	0020_tablenwall_plane_inpainted.jpg	50.7500	57	0057_playground_bench_inpainted.jpg	18.2300
19	0019_tablenwall_giraffe_inpainted.jpg	49.6900	65	0065_beach_bench_inpainted.jpg	19.3300

SSIM (Higher is better)

Best			Worst		
No	INP_FILE	SSIM	No	INP_FILE	SSIM
20	0020_tablenwall_plane_inpainted.jpg	0.9997	63	0063_playground_plane_inpainted.jpg	0.8122
52	0052_tabletop_plane_inpainted.jpg	0.9996	57	0057_playground_bench_inpainted.jpg	0.8384
19	0019_tablenwall_giraffe_inpainted.jpg	0.9996	73	0073_brick_bench_inpainted.jpg	0.8415

LPIPS (Lower is better)

Best			Worst		
No	INP_FILE	LPIPS	No	INP_FILE	LPIPS
56	0056_tabletop_mug_inpainted.jpg	0.0003	63	0063_playground_plane_inpainted.jpg	0.1065
52	0052_tabletop_plane_inpainted.jpg	0.0005	57	0057_playground_bench_inpainted.jpg	0.0886
20	0020_tablenwall_plane_inpainted.jpg	0.0006	73	0073_brick_bench_inpainted.jpg	0.0768

Masked-Region

MASKED_PSNR (Higher is better)

Best			Worst		
No	INP_FILE	MASKED_PSNR	No	INP_FILE	MASKED_PSNR
52	0052_tabletop_plane_inpainted.jpg	43.6000	63	0063_playground_plane_inpainted.jpg	11.9
56	0056_tabletop_mug_inpainted.jpg	42.4500	57	0057_playground_bench_inpainted.jpg	12.0
20	0020_tablenwall_plane_inpainted.jpg	42.0400	65	0065_beach_bench_inpainted.jpg	13.0

MASKED_SSIM (Higher is better)

Best			Worst		
No	INP_FILE	MASKED_SSIM	No	INP_FILE	MASKED_SSIM
52	0052_tabletop_plane_inpainted.jpg	0.9930	63	0063_playground_plane_inpainted.jpg	0.9930
19	0019_tablenwall_giraffe_inpainted.jpg	0.9922	57	0057_playground_bench_inpainted.jpg	0.9930
56	0056_tabletop_mug_inpainted.jpg	0.9918	77	0077_brick_car_inpainted.jpg	0.9930

No-Reference

NIQE (Lower is better)

Best			Worst		
No	INP_FILE	NIQE	No	INP_FILE	NIQE
45	0045_street_car_inpainted.jpg	3.5222	39	0039_sunset_ronaldo_inpainted.jpg	25.1066
48	0048_street_mug_inpainted.jpg	3.5714	40	0040_sunset_mug_inpainted.jpg	24.6295
44	0044_street_plane_inpainted.jpg	3.6393	36	0036_sunset_plane_inpainted.jpg	23.9531

BRISQUE (Lower is better)

Best			Worst		
No	INP_FILE	BRISQUE	No	INP_FILE	BRISQUE
44	0044_street_plane_inpainted.jpg	2.2132	36	0036_sunset_plane_inpainted.jpg	100.9161
48	0048_street_mug_inpainted.jpg	2.2612	34	0034_sunset_tree_inpainted.jpg	98.6659
45	0045_street_car_inpainted.jpg	2.7634	35	0035_sunset_giraffe_inpainted.jpg	98.4069

PIQE (Lower is better)

Best			Worst		
No	INP_FILE	PIQE	No	INP_FILE	PIQE
40	0040_sunset_mug_inpainted.jpg	16.3242	29	0029_wall-floor_car_inpainted.jpg	83.1776
37	0037_sunset_car_inpainted.jpg	19.3444	27	0027_wall-floor_giraffe_inpainted.jpg	82.6826
39	0039_sunset_ronaldo_inpainted.jpg	20.3060	28	0028_wall-floor_plane_inpainted.jpg	82.2695

Benchmarking Report: MIGAN (GPU)

Full-Reference Metrics - Averages

METRIC	AVERAGE
PSNR	35.1328
SSIM	0.9491
LPIPS	0.0204

Combined Full-Reference (formula): $0.4 \cdot \text{norm}(\text{PSNR}) + 0.4 \cdot \text{norm}(\text{SSIM}) + 0.2 \cdot \text{norm_inv}(\text{LPIPS})$

Best Combined

No	INP_FILE	COMBINED_FULLREF
52	0052_tabletop_plane_inpainted.jpg	0.9996
20	0020_tablenwall_plane_inpainted.jpg	0.9990
19	0019_tablenwall_giraffe_inpainted.jpg	0.9972

Worst Combined

No	INP_FILE	COMBINED_FULLREF
63	0063_playground_plane_inpainted.jpg	0.0463
57	0057_playground_bench_inpainted.jpg	0.0820
65	0065_beach_bench_inpainted.jpg	0.1171

Masked-Region Metrics - Averages

METRIC	AVERAGE
masked PSNR	27.8871
masked SSIM	0.7685

Combined Masked (formula): $\text{mean}(\text{norm}(\text{Masked PSNR}), \text{norm}(\text{Masked SSIM}))$

Best Combined

No	INP_FILE	COMBINED_MASKED
52	0052_tabletop_plane_inpainted.jpg	1.0000
56	0056_tabletop_mug_inpainted.jpg	0.9856
20	0020_tablenwall_plane_inpainted.jpg	0.9828

Worst Combined

No	INP_FILE	COMBINED_MASKED
63	0063_playground_plane_inpainted.jpg	0.0381
61	0061_playground_car_inpainted.jpg	0.1086
65	0065_beach_bench_inpainted.jpg	0.1088

No-Reference Metrics - Averages

METRIC	AVERAGE
niqe	7.5256
brisque	41.5878
piqe	50.1701

Combined No-Reference (formula): $\text{mean}(\text{norm_inv}(\text{NIQE}), \text{norm_inv}(\text{BRISQUE}), \text{norm_inv}(\text{PIQE}))$

Best Combined

No	INP_FILE	COMBINED_NOREF
48	0048_street_mug_inpainted.jpg	0.9571
45	0045_street_car_inpainted.jpg	0.9427
44	0044_street_plane_inpainted.jpg	0.9425

Worst Combined

No	INP_FILE	COMBINED_NOREF
35	0035_sunset_giraffe_inpainted.jpg	0.2073
36	0036_sunset_plane_inpainted.jpg	0.2173
40	0040_sunset_mug_inpainted.jpg	0.2766

Top/Bottom Images per Metric

Full-Reference

PSNR (Higher is better)

Best			Worst		
No	INP_FILE	PSNR	No	INP_FILE	PSNR
52	0052_tabletop_plane_inpainted.jpg	50.9100	63	0063_playground_plane_inpainted.jpg	17.5800
20	0020_tablenwall_plane_inpainted.jpg	50.7500	57	0057_playground_bench_inpainted.jpg	18.2300
19	0019_tablenwall_giraffe_inpainted.jpg	49.6900	65	0065_beach_bench_inpainted.jpg	19.3300

SSIM (Higher is better)

Best			Worst		
No	INP_FILE	SSIM	No	INP_FILE	SSIM
20	0020_tablenwall_plane_inpainted.jpg	0.9997	63	0063_playground_plane_inpainted.jpg	0.8122
52	0052_tabletop_plane_inpainted.jpg	0.9996	57	0057_playground_bench_inpainted.jpg	0.8384
19	0019_tablenwall_giraffe_inpainted.jpg	0.9996	73	0073_brick_bench_inpainted.jpg	0.8415

LPIPS (Lower is better)

Best			Worst		
No	INP_FILE	LPIPS	No	INP_FILE	LPIPS
56	0056_tabletop_mug_inpainted.jpg	0.0003	63	0063_playground_plane_inpainted.jpg	0.1065
52	0052_tabletop_plane_inpainted.jpg	0.0005	57	0057_playground_bench_inpainted.jpg	0.0886
20	0020_tablenwall_plane_inpainted.jpg	0.0006	73	0073_brick_bench_inpainted.jpg	0.0768

Masked-Region

MASKED_PSNR (Higher is better)

Best			Worst		
No	INP_FILE	MASKED_PSNR	No	INP_FILE	MASKED_PSNR
52	0052_tabletop_plane_inpainted.jpg	43.6000	63	0063_playground_plane_inpainted.jpg	11.9
56	0056_tabletop_mug_inpainted.jpg	42.4500	57	0057_playground_bench_inpainted.jpg	12.0
20	0020_tablenwall_plane_inpainted.jpg	42.0400	65	0065_beach_bench_inpainted.jpg	13.0

MASKED_SSIM (Higher is better)

Best			Worst		
No	INP_FILE	MASKED_SSIM	No	INP_FILE	MASKED_SSIM
52	0052_tabletop_plane_inpainted.jpg	0.9930	63	0063_playground_plane_inpainted.jpg	0.9930
19	0019_tablenwall_giraffe_inpainted.jpg	0.9922	57	0057_playground_bench_inpainted.jpg	0.9930
56	0056_tabletop_mug_inpainted.jpg	0.9918	77	0077_brick_car_inpainted.jpg	0.9930

No-Reference

NIQE (Lower is better)

Best			Worst		
No	INP_FILE	NIQE	No	INP_FILE	NIQE
45	0045_street_car_inpainted.jpg	3.4246	39	0039_sunset_ronaldo_inpainted.jpg	23.3308
44	0044_street_plane_inpainted.jpg	3.4862	40	0040_sunset_mug_inpainted.jpg	23.3213
48	0048_street_mug_inpainted.jpg	3.4885	36	0036_sunset_plane_inpainted.jpg	23.0031

BRISQUE (Lower is better)

Best			Worst		
No	INP_FILE	BRISQUE	No	INP_FILE	BRISQUE
48	0048_street_mug_inpainted.jpg	2.1809	36	0036_sunset_plane_inpainted.jpg	100.9161
44	0044_street_plane_inpainted.jpg	2.1950	34	0034_sunset_tree_inpainted.jpg	98.6659
45	0045_street_car_inpainted.jpg	2.5052	35	0035_sunset_giraffe_inpainted.jpg	98.4069

PIQE (Lower is better)

Best			Worst		
No	INP_FILE	PIQE	No	INP_FILE	PIQE
40	0040_sunset_mug_inpainted.jpg	16.5921	29	0029_wall-floor_car_inpainted.jpg	82.5042
37	0037_sunset_car_inpainted.jpg	19.4674	28	0028_wall-floor_plane_inpainted.jpg	81.6560
39	0039_sunset_ronaldo_inpainted.jpg	19.9880	27	0027_wall-floor_giraffe_inpainted.jpg	81.3934

Benchmarking Report: AOT-GAN (NPU)

Full-Reference Metrics - Averages

METRIC	AVERAGE
PSNR	34.4822
SSIM	0.9662
LPIPS	0.0214

Combined Full-Reference (formula): $0.4 \cdot \text{norm}(\text{PSNR}) + 0.4 \cdot \text{norm}(\text{SSIM}) + 0.2 \cdot \text{norm_inv}(\text{LPIPS})$

Best Combined

No	INP_FILE	COMBINED_FULLREF
52	0052_tabletop_plane_inpainted.png	0.9984
56	0056_tabletop_mug_inpainted.png	0.9754
20	0020_tablenwall_plane_inpainted.png	0.9634

Worst Combined

No	INP_FILE	COMBINED_FULLREF
63	0063_playground_plane_inpainted.png	0.0528
57	0057_playground_bench_inpainted.png	0.0673
65	0065_beach_bench_inpainted.png	0.1274

Masked-Region Metrics - Averages

METRIC	AVERAGE
masked PSNR	25.5917
masked SSIM	0.7822

Combined Masked (formula): $\text{mean}(\text{norm}(\text{Masked PSNR}), \text{norm}(\text{Masked SSIM}))$

Best Combined

No	INP_FILE	COMBINED_MASKED
56	0056_tabletop_mug_inpainted.png	0.9942
52	0052_tabletop_plane_inpainted.png	0.9701
26	0026_wall-floor_tree_inpainted.png	0.9575

Worst Combined

No	INP_FILE	COMBINED_MASKED
63	0063_playground_plane_inpainted.png	0.0354
65	0065_beach_bench_inpainted.png	0.0760
61	0061_playground_car_inpainted.png	0.1065

No-Reference Metrics - Averages

METRIC	AVERAGE
niqe	5.7254
brisque	37.1004
piqe	41.9427

Combined No-Reference (formula): $\text{mean}(\text{norm_inv}(\text{NIQE}), \text{norm_inv}(\text{BRISQUE}), \text{norm_inv}(\text{PIQE}))$

Best Combined

No	INP_FILE	COMBINED_NOREF
17	0017_tablenwall_bench_inpainted.png	0.9421
16	0016_empty_room_mug_inpainted.png	0.9419
18	0018_tablenwall_tree_inpainted.png	0.9404

Worst Combined

No	INP_FILE	COMBINED_NOREF
39	0039_sunset_ronaldo_inpainted.png	0.2872
40	0040_sunset_mug_inpainted.png	0.3541
36	0036_sunset_plane_inpainted.png	0.4140

Top/Bottom Images per Metric

Full-Reference

PSNR (Higher is better)

Best			Worst		
No	INP_FILE	PSNR	No	INP_FILE	PSNR
52	0052_tabletop_plane_inpainted.png	53.6400	63	0063_playground_plane_inpainted.png	17.8900
56	0056_tabletop_mug_inpainted.png	53.0700	57	0057_playground_bench_inpainted.png	18.0600
20	0020_tablenwall_plane_inpainted.png	50.8100	65	0065_beach_bench_inpainted.png	19.3400

SSIM (Higher is better)

Best			Worst		
No	INP_FILE	SSIM	No	INP_FILE	SSIM
52	0052_tabletop_plane_inpainted.png	0.9991	63	0063_playground_plane_inpainted.png	0.8228
56	0056_tabletop_mug_inpainted.png	0.9990	57	0057_playground_bench_inpainted.png	0.8351
20	0020_tablenwall_plane_inpainted.png	0.9984	73	0073_brick_bench_inpainted.png	0.8524

LPIPS (Lower is better)

Best			Worst		
No	INP_FILE	LPIPS	No	INP_FILE	LPIPS
56	0056_tabletop_mug_inpainted.png	0.0003	63	0063_playground_plane_inpainted.png	0.0984
52	0052_tabletop_plane_inpainted.png	0.0004	57	0057_playground_bench_inpainted.png	0.0886
26	0026_wall-floor_tree_inpainted.png	0.0007	65	0065_beach_bench_inpainted.png	0.0768

Masked-Region

MASKED_PSNR (Higher is better)

Best		
No	INP_FILE	MASKED_PSNR
56	0056_tabletop_mug_inpainted.png	41.5100
52	0052_tabletop_plane_inpainted.png	41.0500
26	0026_wall-floor_tree_inpainted.png	38.6500

Worst		
No	INP_FILE	MASKED_PSNR
63	0063_playground_plane_inpainted.png	11.2300
57	0057_playground_bench_inpainted.png	11.3900
65	0065_beach_bench_inpainted.png	12.3900

MASKED_SSIM (Higher is better)

Best		
No	INP_FILE	MASKED_SSIM
56	0056_tabletop_mug_inpainted.png	0.9880
26	0026_wall-floor_tree_inpainted.png	0.9859
52	0052_tabletop_plane_inpainted.png	0.9850

Worst		
No	INP_FILE	MASKED_SSIM
63	0063_playground_plane_inpainted.png	0.2850
57	0057_playground_bench_inpainted.png	0.3580
77	0077_brick_car_inpainted.png	0.4070

No-Reference

NIQE (Lower is better)

Best		
No	INP_FILE	NIQE
20	0020_tablenwall_plane_inpainted.png	4.1037
43	0043_street_giraffe_inpainted.png	4.1485
24	0024_tablenwall_mug_inpainted.png	4.1678

Worst		
No	INP_FILE	NIQE
40	0040_sunset_mug_inpainted.png	17.5188
39	0039_sunset_ronaldo_inpainted.png	16.5957
36	0036_sunset_plane_inpainted.png	16.5873

BRISQUE (Lower is better)

Best		
No	INP_FILE	BRISQUE
16	0016_empty_room_mug_inpainted.png	16.5054
48	0048_street_mug_inpainted.png	16.8924
18	0018_tablenwall_tree_inpainted.png	17.0620

Worst		
No	INP_FILE	BRISQUE
36	0036_sunset_plane_inpainted.png	98.6659
35	0035_sunset_giraffe_inpainted.png	96.0028
40	0040_sunset_mug_inpainted.png	92.4862

PIQE (Lower is better)

Best		
No	INP_FILE	PIQE
39	0039_sunset_ronaldo_inpainted.png	17.5929
40	0040_sunset_mug_inpainted.png	18.2323
38	0038_sunset_book_inpainted.png	21.3653

Worst		
No	INP_FILE	PIQE
29	0029_wall-floor_car_inpainted.png	73.1970
28	0028_wall-floor_plane_inpainted.png	72.8465
31	0031_wall-floor_ronaldo_inpainted.png	71.9723

Benchmarking Report: LaMa (NPU)

Full-Reference Metrics - Averages

METRIC	AVERAGE
PSNR	36.7151
SSIM	0.9779
LPIPS	0.0186

Combined Full-Reference (formula): $0.4 \cdot \text{norm}(\text{PSNR}) + 0.4 \cdot \text{norm}(\text{SSIM}) + 0.2 \cdot \text{norm_inv}(\text{LPIPS})$

Best Combined

No	INP_FILE	COMBINED_FULLREF
20	0020_tablenwall_plane_inpainted.png	0.9930
52	0052_tabletop_plane_inpainted.png	0.9880
56	0056_tabletop_mug_inpainted.png	0.9750

Worst Combined

No	INP_FILE	COMBINED_FULLREF
70	0070_beach_book_inpainted.png	0.0463
73	0073_brick_bench_inpainted.png	0.1219
57	0057_playground_bench_inpainted.png	0.1332

Masked-Region Metrics - Averages

METRIC	AVERAGE
masked PSNR	24.6554
masked SSIM	0.7766

Combined Masked (formula): $\text{mean}(\text{norm}(\text{Masked PSNR}), \text{norm}(\text{Masked SSIM}))$

Best Combined

No	INP_FILE	COMBINED_MASKED
56	0056_tabletop_mug_inpainted.png	0.9951
52	0052_tabletop_plane_inpainted.png	0.9926
26	0026_wall-floor_tree_inpainted.png	0.9806

Worst Combined

No	INP_FILE	COMBINED_MASKED
64	0064_playground_mug_inpainted.png	0.0544
77	0077_brick_car_inpainted.png	0.1118
69	0069_beach_car_inpainted.png	0.1466

No-Reference Metrics - Averages

METRIC	AVERAGE
niqe	6.7428
brisque	38.4588
piqe	41.7638

Combined No-Reference (formula): $\text{mean}(\text{norm_inv}(\text{NIQE}), \text{norm_inv}(\text{BRISQUE}), \text{norm_inv}(\text{PIQE}))$

Best Combined

No	INP_FILE	COMBINED_NOREF
17	0017_tablenwall_bench_inpainted.png	0.9413
23	0023_tablenwall_ronaldo_inpainted.png	0.9388
21	0021_tablenwall_car_inpainted.png	0.9348

Worst Combined

No	INP_FILE	COMBINED_NOREF
36	0036_sunset_plane_inpainted.png	0.2079
39	0039_sunset_ronaldo_inpainted.png	0.3013
35	0035_sunset_giraffe_inpainted.png	0.3697

Top/Bottom Images per Metric

Full-Reference

PSNR (Higher is better)

Best			Worst		
No	INP_FILE	PSNR	No	INP_FILE	PSNR
20	0020_tablenwall_plane_inpainted.png	50.1900	70	0070_beach_book_inpainted.png	21.2600
52	0052_tabletop_plane_inpainted.png	49.3300	64	0064_playground_mug_inpainted.png	21.6000
19	0019_tablenwall_giraffe_inpainted.png	49.0300	69	0069_beach_car_inpainted.png	21.9800

SSIM (Higher is better)

Best			Worst		
No	INP_FILE	SSIM	No	INP_FILE	SSIM
52	0052_tabletop_plane_inpainted.png	0.9984	73	0073_brick_bench_inpainted.png	0.9168
56	0056_tabletop_mug_inpainted.png	0.9982	74	0074_brick_tree_inpainted.png	0.9242
26	0026_wall-floor_tree_inpainted.png	0.9982	57	0057_playground_bench_inpainted.png	0.9272

LPIPS (Lower is better)

Best			Worst		
No	INP_FILE	LPIPS	No	INP_FILE	LPIPS
56	0056_tabletop_mug_inpainted.png	0.0005	70	0070_beach_book_inpainted.png	0.0721
52	0052_tabletop_plane_inpainted.png	0.0009	73	0073_brick_bench_inpainted.png	0.0573
20	0020_tablenwall_plane_inpainted.png	0.0016	74	0074_brick_tree_inpainted.png	0.0515

Masked-Region

MASKED_PSNR (Higher is better)

Best		
No	INP_FILE	MASKED_PSNR
52	0052_tabletop_plane_inpainted.png	38.8300
56	0056_tabletop_mug_inpainted.png	38.6400
26	0026_wall-floor_tree_inpainted.png	38.3000

Worst		
No	INP_FILE	MASKED_PSNR
64	0064_playground_mug_inpainted.png	9.990
70	0070_beach_book_inpainted.png	10.790
58	0058_playground_tree_inpainted.png	11.140

MASKED_SSIM (Higher is better)

Best		
No	INP_FILE	MASKED_SSIM
12	0012_empty_room_plane_inpainted.png	0.9873
56	0056_tabletop_mug_inpainted.png	0.9856
10	0010_empty_room_tree_inpainted.png	0.9849

Worst		
No	INP_FILE	MASKED_SSIM
77	0077_brick_car_inpainted.png	0.2599
64	0064_playground_mug_inpainted.png	0.3440
74	0074_brick_tree_inpainted.png	0.4065

No-Reference

NIQE (Lower is better)

Best		
No	INP_FILE	NIQE
48	0048_street_mug_inpainted.png	3.4283
41	0041_street_bench_inpainted.png	3.4916
47	0047_street_ronaldo_inpainted.png	3.5135

Worst		
No	INP_FILE	NIQE
36	0036_sunset_plane_inpainted.png	28.5307
39	0039_sunset_ronaldo_inpainted.png	25.6882
35	0035_sunset_giraffe_inpainted.png	23.6335

BRISQUE (Lower is better)

Best		
No	INP_FILE	BRISQUE
24	0024_tablenwall_mug_inpainted.png	0.0573
18	0018_tablenwall_tree_inpainted.png	0.2238
17	0017_tablenwall_bench_inpainted.png	0.2647

Worst		
No	INP_FILE	BRISQUE
36	0036_sunset_plane_inpainted.png	97.4691
35	0035_sunset_giraffe_inpainted.png	96.2219
40	0040_sunset_mug_inpainted.png	96.0173

PIQE (Lower is better)

Best		
No	INP_FILE	PIQE
40	0040_sunset_mug_inpainted.png	17.6534
38	0038_sunset_book_inpainted.png	18.6756
34	0034_sunset_tree_inpainted.png	19.6481

Worst		
No	INP_FILE	PIQE
29	0029_wall-floor_car_inpainted.png	74.2088
31	0031_wall-floor_ronaldo_inpainted.png	73.3444
28	0028_wall-floor_plane_inpainted.png	71.5545